higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

AUTOMOTIVE REPAIR AND MAINTENANCE
NQF LEVEL 4

12 MARCH 2019

This marking guideline consists of 5 pages.

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QUESTION 1

1.1 1.1.1 B
1.1.2 A
1.1.3 B
1.1.4 B
1.1.5 B

(5 × 1) (5)

1.2 A – Pinion gear
B – Cross shaft/carrier shaft
C – Planetary carrier
D – Bearing
E – Casing/housing
F – sun/side gear
G – Planetary gear
H – Crown wheel

(8 × 1) (8)

1.3 **Disadvantages:**

- Uses engine power to be driven
- Broken fan belt and lack of oil can cause hard steering
- Increased maintenance
- Increased weight, complication and cost

Advantages:

- Minimises driver fatigue
- Reduces the number of turns from lock to lock
- Lessens kickback at the steering wheel
- Imposes safety by better resisting any sudden swerving of the vehicle

(Any 7 × 1) (7)
[20]

QUESTION 2

2.1 2.1.1 L
2.1.2 D
2.1.3 I
2.1.4 H
2.1.5 M
2.1.6 G
2.1.7 C
2.1.8 A
2.1.9 B
2.1.10 E

(10 × 1) (10)

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2.2 **Advantages:**

- Various gear ratios may be obtained.
- More torque can be transmitted.
- Limited wear takes place.
- It provides a quiet drive.
- Gears can be changed without interrupting the transmission of torque.

(Any THREE)

Disadvantages:

- It is more expensive than the conventional.
- Brake bands that can wear are used.
- Additional elements must be used to make it more versatile. (THREE)

(3 + 3) (6)

2.3

- Sun gear
- Planetary gear
- Planet carrier
- Annulus or ring gear (4)

[20]

QUESTION 3

3.1 3.1.1 False
3.1.2 True
3.1.3 True
3.1.4 False
3.1.5 False

(5 × 1) (5)

3.2 3.2.1 Swept volume is the volume swept through in the cylinder during one stroke and is equal to the cross-sectional piston area multiplied by the stoke length✓. Volumetric efficiency is the ability of the engine to fill the engine above the piston at the end of the induction stroke. ✓

3.2.2 A solenoid converts electrical energy to mechanical energy✓. A sensor sends a signal to an electric control unit (ECH). ✓

3.2.3 Mechanical efficiency is the effectiveness of the engine during operation. ✓ Thermal efficiency is the heat found by comparing the burned fuel and the output of the engine in kilowatt. ✓

3.2.4 Static crankshaft balancing is the equal distribution of the mass about its axis of rotation. ✓ Dynamic crankshaft balancing means an assembly must rotate over its speed range on an axis that runs through the centre of the assembly at the same time. ✓

3.2.5 A compression test tests the compression developed in the combustion chamber. ✓ A cylinder-balance test checks which cylinder is under performing. ✓

(5 × 2) (10)

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3.3 It consists of an ordinary pinion gear and crown wheel✓ giving it the first primary reduction.✓ The helical pinion gear is integrally mounted in the cross shaft✓ of the crown wheel and meshes with the helical drive gear✓ to form the second reduction.✓ (5)
[20]

QUESTION 4

4.1
4.1.1 decreases
4.1.2 volumetric
4.1.3 pressure
4.1.4 less
4.1.5 higher
(5 × 1) (5)

4.2 **Advantages:**
- Engine power is increased.
- More kilowatts may be obtained without increasing the engine size.
- It has a longer life span.
- It is reliable.
- It has a small fire hazard.
- It has a high combustion rate.
- Fuel consumption is lower because of high thermal efficiency.
- No interference with radio signals
- Maintenance cost is low because of longer intervals between services.

(Any FIVE)

Disadvantages:
- It is more expensive.
- Starting aids are often required.
- It has a lower acceleration.
- Construction is heavier.
- It is noisy. (FIVE)

(5 + 5) (10)

4.3 It is the heat found by comparing the burned fuel and the output of the engine in kilowatt.✓ The thermal efficiency for petrol is 30%✓and 40% for diesel.✓ It indicates how well the engine can use the fuel heat energy.✓ It measures the ability of the engine to convert as much heat as possible to work.✓ (5)
[20]

QUESTION 5

5.1 The heat is taken from the combustion of the fuel✓ where there is very little movement if the piston at TDC.✓ Heat is expelled from the cylinder if the exhaust valve opens.✓ The heat exchange operations occur while the volume of the gases in the cylinder remains constant.✓ This is called the constant volume.✓ (5)

5.2 To increase the ratio, the combustion chamber volume is made smaller by:
- Fitting a high-compression cylinder head gasket
- Skimming the cylinder head
- Using pistons that protrude into the combustion chamber
- Grounding the crankshaft journals off-centre
- Fitting longer connecting rods (5)

5.3
- Warm up the engine.
- Remove spark plugs.
- Remove air cleaners.
- Screw an external air hose coupling into no. 1 plug hole.
- Turn engine to no. 1 piston TDC.
- Force external air into the cylinder and observe the gauge.
- Repeat the process on other cylinders. (Any 5 × 1) (5)

5.4 5.4.1 Investigate and apply safe working practices throughout the task.

5.4.2 Remove and replace the faulty electronic ignition system component with parts that meet the manufacturer's specifications.

5.4.3 Test the system to ensure that the integrity of the wiring is in a serviceable condition.

5.4.4 Check all adjustments and, where necessary, correct to comply with the manufacturer's specifications.

5.4.5 Confirm if the electronic ignition system operates according to the manufacturer's specifications.

(5 × 1) (5)
[20]

TOTAL: 100

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